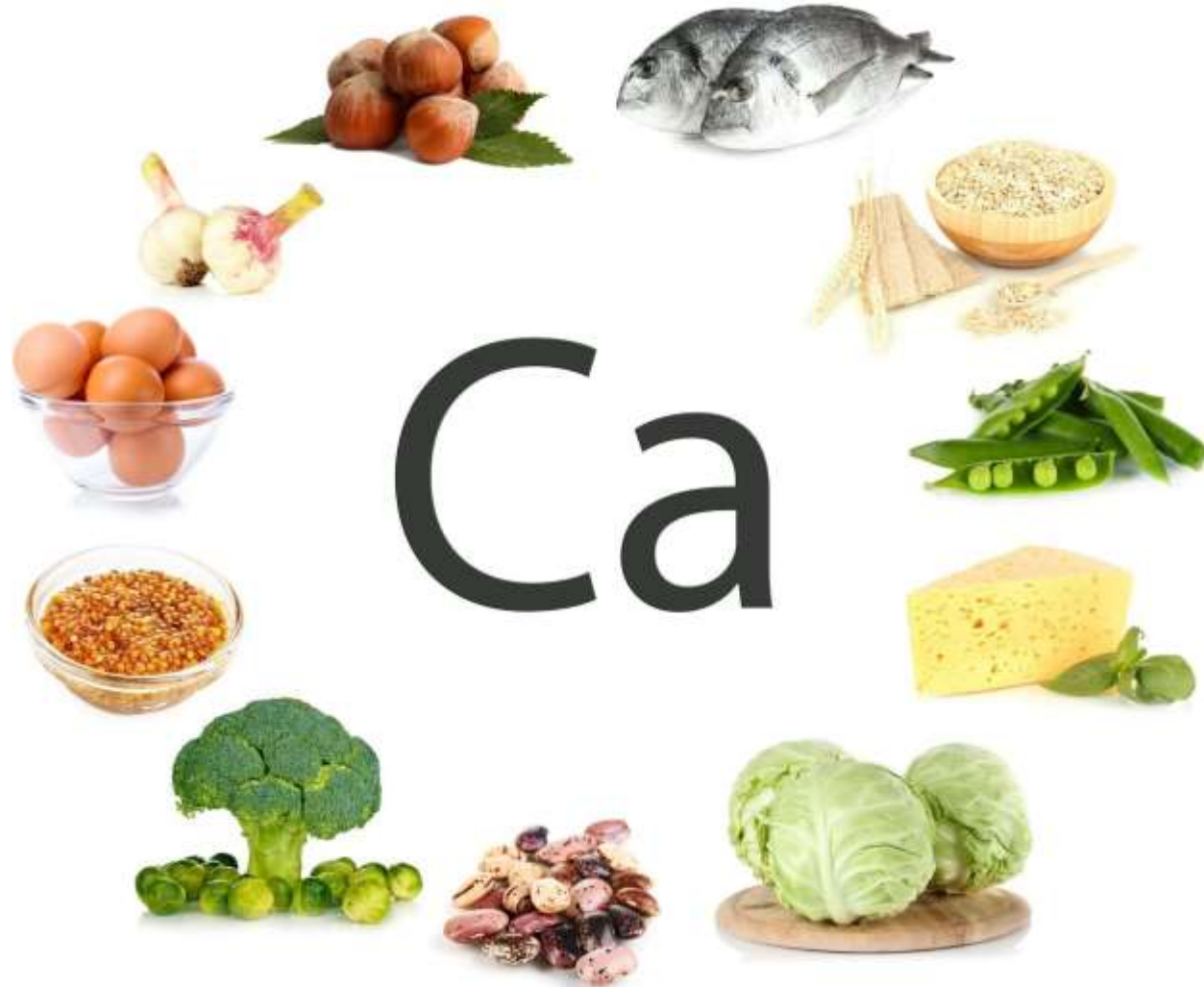


Calcium



Calcium Ca^{2+}

- Serum level is 4.5-5mEq/L.
- Most abundant mineral in the human body.
- The average adult body contains 1-1.4kg, 99% is in teeth and bones.

- Plasma Ca exists in 3 forms, ionized Ca^{2+} , protein bounded(albumin), and complex form as(citrate, phosphate, and bicarbonate).
- needed for nerve transmission, vitamin B12 absorption, muscle contraction & blood clotting

Hypocalcaemia:

- Results from decreased intake as in
 - Vitamin D3 deficiency, lack of sunlight plus poor nutrition, and malabsorption.
- Decreased plasma albumin as in malnutrition, & burns.
- Increased renal loss by calcitonin hormone.
- Parathyroid disorders.

Hypocalcaemia, cont.

- Decreased flux from bone as in PTH deficiency, & surgical removal of glands.
- Increased flux into bone “hungry bones” as in postparathyroidectomy.
- Symptoms of hypocalcaemia: tingling of the finger tips; abdominal & muscle cramps; tetany; seizures; osteomalacia; & rickets.

Hypercalcaemia:

- Results from increased intake/absorption e.g. vitamin D3.
- Increased plasma albumin as in dehydration.
- Increased bone resorption as in hyperparathyroidism, & malignancy.

Hypercalcaemia, cont.

- Increased renal reabsorption as in thiazide diuretics, & Addison's disease.
- Symptoms are fatigue, depression, anorexia, muscle weakness, renal calculi, & cardiac changes.

Functions of calcium

- Building strong bones and teeth.
- Clotting blood.
- Sending and receiving nerve signals.
- Squeezing and relaxing muscles.
- Releasing hormones and other chemicals.
- Keeping a normal heartbeat.

Sources of calcium

- milk
- yoghurt and cheese
- calcium fortified foods (such as soy products)
- some leafy green vegetables
- nuts and seeds